Label Degeneracy and Label Leakage in Learned Quantum Error Mitigation: A Corrected Evaluation of a Hybrid ML+VQE Pipeline for Grover Search

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Abstract

Learned error mitigation trains a model to map noisy quantum measurement outcomes to the ideal outcomes, and it can look highly effective when evaluated under weak protocols. We revisit a hybrid pipeline that combines a random forest (RF) and a multilayer perceptron (MLP) with a variational quantum eigensolver (VQE) probe feature to mitigate depolarizing and amplitude-damping noise in Grover search circuits. An earlier version of the pipeline reported fidelity rising from 0.74 to 0.99 and marked-state probability from 0.05 to 0.48. We show that these results rest on two methodological defects. First, because the training circuit and the simulator seed were fixed, every training label was the same vector, so a model could fit a constant target without learning a noise-correction map. Second, the ensemble blend weight used the ground-truth ideal distribution at inference, which leaks the target. The VQE feature was also computed noiselessly, so it could not carry noise information. We fix these defects, introduce a held-out-target protocol with ten training targets and three unseen test targets, repeat evaluation over 20 shot-noise seeds per target, and add an ablation of the feature set and model class. Under the corrected protocol the hybrid pipeline does not outperform simple zero-noise extrapolation (ZNE). At four qubits, ZNE reaches fidelity 0.879 ± 0.017 against 0.680 ± 0.011 for the best hybrid variant that includes VQE and 0.690 ± 0.005 for the best overall learned variant. The unmitigated noisy output reaches 0.758 ± 0.026, so the learned pipeline is worse than no mitigation on fidelity. A five-qubit check shows the same ordering with a smaller margin. We argue that the main contribution is methodological: label degeneracy and label leakage can inflate reported gains in learned error mitigation, and a corrected multi-target protocol is required to detect them.

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1 Introduction

Near-term quantum processors are limited by noise, and noise degrades the output of even short algorithms. Grover's search [3] is a natural benchmark because its success probability depends sharply on how cleanly the amplified state is measured. Error mitigation aims to recover useful output from noisy circuits without the overhead of full quantum error correction [8, 2]. Two families are common. Model-based methods extrapolate noisy measurements to a zero-noise limit, with zero-noise extrapolation (ZNE) being the most widely used [8, 4]. Learning-based methods train a model, usually a regressor or a classifier, to map noisy outputs to ideal outputs. Learned methods are attractive because they can exploit regularities in the noise and the circuit, but they are also easy to evaluate badly.

This paper concerns a specific learned pipeline. A random forest [1] and a multilayer perceptron [7] predict the ideal output distribution from a noisy histogram, optionally augmented with two scalar features: the noise level and a VQE-derived probe [5]. The original submission reported large improvements, with fidelity rising from 0.74 to 0.99 and marked-state probability from 0.05 to 0.48. Re-examining the implementation shows that these numbers are not evidence that the method works.

The contributions of this paper are:

(i) We identify and demonstrate a *label-degeneracy* defect. In the original design every training sample shares one ideal target, so the set of training labels has exactly one distinct value.

(ii) We identify a *label-leakage* defect. The original ensemble blend weight is a fidelity computed against the true ideal distribution, which is unavailable at inference time.

(iii) We identify a *noise-independence* defect in the VQE feature, which is computed without noise and therefore constant across noise levels.

(iv) We fix all three defects and evaluate on held-out marked states, with 20 shot-noise seeds per target, reporting mean and standard deviation.

(v) We report that under this corrected protocol the learned pipeline does not outperform ZNE at four or five qubits, and that at four qubits it is worse than the unmitigated output on fidelity.

We treat the negative result as the contribution. Learned mitigation is not ruled out in general. What this study shows is that a particular pipeline, evaluated under a weak protocol, appears strong, and that the same pipeline is not competitive once the protocol is corrected. The defects we describe are simple and general, and we expect them to recur in other learned-mitigation studies that train on a single circuit.

The paper is organized as follows. Section 3 gives the background and the noise model. Section 4 describes the original design and its defects in detail. Section 5 gives the corrected pipeline and protocol. Section 6 reports the four-qubit and five-qubit results and the ablation. Section 7 discusses implications, and Section 8 lists threats to validity. Section 10 describes reproducibility, and Section 11 concludes. The appendices give the configuration and the full per-target breakdown.

2 Related Work

2.1 Error mitigation without full correction

Quantum error correction is out of reach for current hardware, so error mitigation aims to reduce the bias of expectation values or output distributions using additional circuit executions and classical post-processing [2]. A recent review surveys the field [9]. Zero-noise extrapolation runs the circuit at several amplified noise scales and extrapolates the result to zero noise [8]. Richardson-type and exponential extrapolations are common variants [11]. ZNE has been demonstrated on superconducting hardware [4] and remains the baseline most mitigation methods are compared against.

2.2 Learned error mitigation

Learned mitigation replaces or augments extrapolation with a supervised model that maps noisy outputs to ideal ones. Such models are trained on pairs of noisy and noiseless executions of training circuits, and their accuracy depends on how well the training pairs cover the space of circuits, noise levels, and target states. This coverage requirement is what makes evaluation delicate. A model trained and tested on the same circuit can achieve low error without learning the noise, and a model trained on one circuit may memorize that circuit's output distribution. Our study exposes this failure mode in a concrete pipeline.

2.3 Variational quantum eigensolver as a probe

The variational quantum eigensolver [5] minimizes an expectation value over a parameterized circuit, within the general framework of variational hybrid algorithms [12, 10]. In this pipeline it serves as a noise-sensitive probe rather than as a solver: the minimal energy found under a given noise model is used as a scalar feature. This use is heuristic. Its value depends on whether the probe's energy varies with noise in a useful way, which is an empirical question that our corrected experiments address.

2.4 Evaluation practice

Reported gains in learned mitigation often come from single circuits, single noise realizations, and single seeds, which makes the variance of the reported quantity unknown. Multi-target evaluation, with training and test targets drawn from disjoint sets, is the standard guard against memorization in supervised learning. Reporting variance over seeds is the standard guard against over-interpreting a single run. Our corrected protocol applies both.

3 Background

3.1 Grover search circuits

A single-iteration Grover circuit on n qubits prepares a uniform superposition with Hadamard gates, applies an oracle that flips the phase of a marked bitstring, applies the diffusion operator,

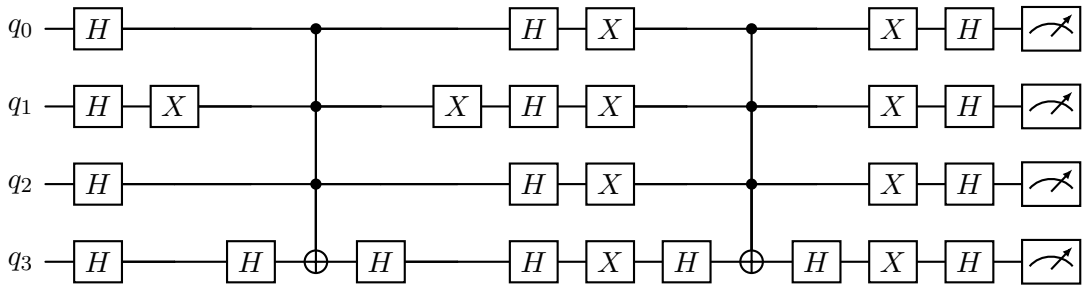


Figure 1: Single-iteration Grover circuit for the marked state `1011` on four qubits, as built by `create_grover(4, "1011")`. Column 1 prepares the uniform superposition, columns 2–6 implement the oracle (the first X on q_1, the H–$\mathrm{C^3X}$–H block on q_3, and the second X on q_1), and columns 7–13 implement the diffusion operator. The circuit is simulated with 2048 shots, and every qubit is measured at the end.

and measures in the computational basis [3]. For $n = 4$ the ideal output concentrates probability on the marked bitstring, and the pipeline's job is to recover that concentration from noisy measurements. The repository builds one circuit per target with `create_grover(n_qubits, target)`, which generalizes the original four-qubit construction to other marked states and to five qubits.

Figure 1 shows the four-qubit circuit for the marked state `1011`, with the gate sequence exactly as `create_grover` emits it. The circuit prepares a uniform superposition, applies an oracle that flips the phase of the marked state, applies the diffusion operator, and measures all qubits in the computational basis. Qubit q_i corresponds to the i-th character of the target string, so the two X gates on q_1 conjugate the multi-controlled Toffoli so that it acts only on the basis state `1011`. Because the circuit is a single iteration on $N = 16$ states, the ideal probability of measuring the marked state is $\sin^2(3\theta) \approx 0.47$ with $\theta = \arcsin(1/4)$. This value is the reference against which the noisy and mitigated outputs in Section 6 should be read.

3.2 Noise model

Noise is applied through a Qiskit Aer noise model at a single level $\lambda \in [0, 1]$. Each single-qubit gate (`u1`, `u2`, `u3`) is followed by a depolarizing error of strength λ and an amplitude-damping error of strength λ. The noisy output is the histogram of $S = 2048$ measurement shots. The model is a simulator model: it is not calibrated to any hardware device, and it omits readout error, two-qubit gate error, crosstalk, and drift. The results therefore describe mitigation under this noise model and should not be read as device results.

3.3 Output representation

The histogram is converted to a probability vector of length 2^n (16 entries at four qubits, 32 at five) using `counts_to_vector`. Ideal targets are the noiseless output vectors of the same circuits, computed with the same shot budget and a different seed. Accuracy of a mitigated vector is evaluated against the ideal vector with three metrics. Fidelity is $F(p, q) = \left(\sum_i \sqrt{p_i q_i}\right)^2$, where higher is better. KL divergence is $D_{\mathrm{KL}}(p\|q) = \sum_i p_i \log\left((p_i + 10^{-10})/(q_i + 10^{-10})\right)$ with a small smoothing constant, where lower is better. Mean squared error is $\frac{1}{2^n}\sum_i (p_i - q_i)^2$, where lower is

better.

3.4 Zero-noise extrapolation baseline

The ZNE baseline runs the circuit at noise scales 0.05, 0.10, and 0.15, each with its own seed offset, and linearly extrapolates the first two outputs to zero noise as $\hat{p} = 2r_{0.05} - r_{0.10}$. The third scale is computed but not used in the extrapolation. This is the simplest form of the method. More elaborate extrapolations, such as higher-order polynomial or exponential fits and the third scale, could change the ZNE numbers, and we return to this in Section 8.

3.5 Variational probe

The VQE probe uses a two-qubit hardware-efficient ansatz (`TwoLocal` with `ry` and `rz` rotations and `cz` entanglement, two repetitions) to minimize the expectation of the Hamiltonian $H = ZZ + IZ$ with COBYLA for 50 iterations and random initialization. The estimator runs with 1,024 shots under the noise model at the matching level. The minimized energy is the scalar VQE feature. Because the probe is run under noise, its value depends on the noise level, which is the property the original design lacked.

4 The Original Design and Its Defects

4.1 Original training data

In the original design, every training sample used the same Grover circuit and the same target bitstring. The simulator was called with a constant seed, so the noiseless "ideal" execution was deterministic. Every training label y was therefore the same vector.

We verified this defect directly. Reconstructing the original training loop and computing the number of distinct label rows with `np.unique(y, axis=0)` returns one row for all 400 training samples. A regressor fitted to a constant target can reach near-zero training error by outputting that constant, without any dependence on the noisy input. Any reported improvement over an unmitigated baseline must then be checked against the trivial solution of outputting the target itself, and the original evaluation did not perform that check.

4.2 Label leakage in the ensemble weight

The original adaptive ensemble set the RF/MLP blend weight from the fidelity between the MLP prediction and the true ideal distribution. The ideal distribution is the quantity the pipeline is supposed to recover. At inference time it is unknown, so a weight computed from it is not available to a deployed model. Using it in the evaluation therefore lets the reported output depend on the answer, which inflates the reported numbers.

4.3 Noise-independent VQE feature

The original VQE probe used a noiseless estimator. Its minimized energy is the same at every noise level, so the feature is a constant with respect to λ and cannot carry the noise information the original paper attributed to it. This defect does not by itself inflate the reported gains, but it means the VQE ablation in the original design was not testing what it claimed to test.

4.4 Consequences for the reported result

The three defects combine. The degeneracy makes the training task trivial; the leakage makes the evaluated output depend on the target; and the constant feature removes an input the design depended on. We do not claim to reproduce the original numbers exactly. We claim that the original protocol cannot distinguish a learned noise correction from a memorized constant, and that once the protocol is corrected the reported gains do not appear.

5 Corrected Pipeline and Protocol

5.1 Training targets and circuits

The four-qubit experiment uses ten training targets, $\{0000, 0011, 0101, 0110, 1001, 1010, 1100, 1111, 0001, 1110\}$. One circuit is built for each target. Noise levels for the training set are six values evenly spaced in $[0.02, 0.18]$. Each (target, noise level) cell contains 15 noisy executions, paired with the ideal vector of the same target. The training matrix has shape 900×18, and the label matrix has ten distinct rows.

5.2 Model inputs

Each training input is an 18-dimensional vector: the 16-entry noisy probability vector, the noise level λ used for that execution, and the VQE feature computed at that λ. Each target contributes samples at every noise level, so the models learn a map that depends on the noise level and the noisy histogram.

5.3 Held-out targets

Three test targets, $\{1011, 0100, 1101\}$, are excluded from training. They are never seen in any training sample, so evaluation on them tests generalization across marked states rather than across noise realizations of a memorized circuit. The test noise level is $\lambda = 0.10$, which lies between the training grid points and is therefore also not a training level.

5.4 Shot-noise seeds

For each held-out target we draw 20 independent shot-noise seeds at $\lambda = 0.10$, giving 60 trials per method. Each trial produces a noisy histogram, and each method's output is compared with the ideal vector of the same target. Headline numbers are reported as mean and standard deviation over the 60 trials, and per-target breakdowns are also reported.

5.5 Models

The random forest is a scikit-learn `RandomForestRegressor` with 200 trees, fitted to the 16-dimensional ideal vector. The MLP is an `MLPRegressor` with hidden layers $(128, 64, 32)$, early stopping with a 15% validation fraction, a patience of 25 epochs, and a maximum of 2,000 iterations. Both models use seed 42.

5.6 Leak-free ensemble

The ensemble combines the normalized RF and MLP outputs with a weight that depends only on their mutual agreement. Let r and m be the normalized RF and MLP outputs. The blend weight is $w = F(r, m)$, the fidelity between them, and the output is $\hat{p} = w\, m + (1 - w)\, r$, clipped to be non-negative and renormalized. Because the weight uses only the two model predictions, it is available at inference and does not depend on the true ideal. The rule is a heuristic: agreement between two regressors is a plausible but unverified proxy for reliability.

5.7 Variants

Four learned configurations are compared on the same training and test sets. The RF-only and MLP-only variants use the full 18-dimensional input, including the VQE feature. The ensemble-with-VQE variant uses the full input. The ensemble-without-VQE variant drops the VQE feature and uses the 17-dimensional input. Together these isolate the contribution of the model class and of the probe.

5.8 Baselines

Two baselines are reported. The noisy baseline is the unmitigated histogram at $\lambda = 0.10$. The ZNE baseline is the two-point extrapolation described in Section 3. Both are evaluated on the same 60 trials per target.

5.9 Five-qubit scaling protocol

The five-qubit experiment uses six training targets, four noise levels evenly spaced in $[0.02, 0.18]$, and 10 repetitions per cell, giving 240 samples for a 32-dimensional label. Two held-out targets, $\{10011, 01101\}$, are evaluated with 10 seeds each at $\lambda = 0.10$. The proportionally smaller training set is a deliberate choice that makes the cost of the $O(2^n)$ label space visible. It is also a weaker training set relative to the output dimension, which we discuss in Section 8.

6 Results

6.1 Four-qubit held-out generalization

Table 1 gives the pooled results over three held-out targets and 20 seeds each. Figure 2 displays the same comparison as bar charts with standard-deviation error bars.

Table 1: 4-qubit held-out generalization at $\lambda = 0.10$ (mean $\pm$ standard deviation over 3 targets $\times$ 20 seeds).

Method	Fidelity ↑	KL ↓	MSE ↓
Noisy (no mitigation)	0.758 ± 0.026	0.665 ± 0.098	0.0114 ± 0.0010
ZNE	**0.879 ± 0.017**	**0.272 ± 0.042**	**0.0064 ± 0.0008**
RF only (+VQE)	0.662 ± 0.023	1.005 ± 0.045	0.0146 ± 0.0017
MLP only (+VQE)	0.680 ± 0.011	0.986 ± 0.048	0.0135 ± 0.0004
Ensemble, no VQE	0.690 ± 0.005	0.944 ± 0.026	0.0133 ± 0.0002
Ensemble, +VQE	0.680 ± 0.011	0.986 ± 0.047	0.0135 ± 0.0004

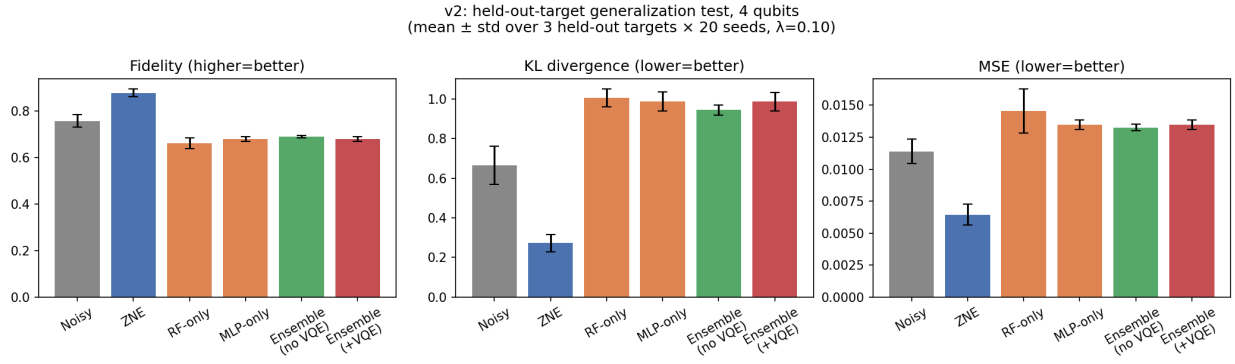


Figure 2: 4-qubit held-out comparison. Bars show means and error bars show standard deviations over 3 targets $\times$ 20 seeds. Generated from the repository's saved run.

ZNE is best on every metric. Its fidelity exceeds the best learned variant by 0.189, and its KL and MSE are roughly a quarter and a half of the learned values, respectively. The unmitigated noisy output has higher fidelity than every learned variant and lower KL and MSE than every learned variant. In this sense the learned pipeline does not improve on the input it was designed to correct. Its only comparative advantage is variance: the ensemble-without-VQE variant has the smallest fidelity standard deviation (0.005), which reflects stability rather than accuracy.

6.2 Per-target consistency

Table 2 gives the fidelity breakdown by held-out target. The ranking is the same for each target. ZNE leads on every target, the noisy baseline is second on fidelity, and the learned variants are below it on all three targets. The learned variants also cluster closely, so no single learned variant is clearly superior on every target.

The per-target table also shows that the variance across targets is smaller than the gap between ZNE and the learned variants. The target `1101` is the hardest for the noisy baseline, ZNE, and every learned variant, which is consistent with a dependence of mitigation quality on the marked state.

6.3 Ablation

Dropping the VQE feature from the ensemble improves every pooled metric (fidelity 0.690 versus 0.680; KL 0.944 versus 0.986; MSE 0.0133 versus 0.0135). On fidelity the improvement holds on

Table 2: Fidelity by held-out target at $\lambda = 0.10$ (mean ± standard deviation over 20 seeds).

Method	0100	1011	1101
Noisy	0.792 ± 0.012	0.744 ± 0.010	0.740 ± 0.010
ZNE	**0.894 ± 0.011**	**0.874 ± 0.015**	**0.869 ± 0.012**
RF only (+VQE)	0.672 ± 0.007	0.665 ± 0.021	0.650 ± 0.031
MLP only (+VQE)	0.692 ± 0.002	0.680 ± 0.004	0.667 ± 0.003
Ensemble, no VQE	0.693 ± 0.002	0.694 ± 0.002	0.683 ± 0.003
Ensemble, +VQE	0.692 ± 0.002	0.680 ± 0.004	0.667 ± 0.002

each held-out target individually; on KL it does not hold for target 0100, where the two ensembles are within 0.01 of each other. RF alone is the weakest single model (fidelity 0.662, KL 1.005). The MLP and the ensemble are close.

The ablation supports a cautious reading. The VQE feature, once computed under noise, does not add signal under this training budget and adds variance. Two explanations are consistent with the data. The first is that the 18-dimensional representation with roughly 900 training samples is too small for the models to learn a transferable correction across marked states. The second is that the VQE probe, run with COBYLA for 50 iterations and random restarts, is too noisy a feature to be useful at this scale. We cannot distinguish these explanations with the present experiments.

6.4 Five-qubit scaling check

Table 3 and Figure 3 give the five-qubit results. ZNE and the noisy output are nearly tied, with ZNE slightly ahead on fidelity and KL, and the hybrid ensemble trails both on all three metrics. The noisy and ZNE fidelities are 0.852 ± 0.008 and 0.858 ± 0.015, a gap of 0.006 that is within one standard deviation of either. The hybrid ensemble reaches 0.822 ± 0.009.

Table 3: 5-qubit generalization check at $\lambda = 0.10$ (mean ± standard deviation over 2 targets × 10 seeds).

Method	Fidelity ↑	KL ↓	MSE ↓
Noisy	0.852 ± 0.008	0.421 ± 0.030	0.0019 ± 0.0001
ZNE	**0.858 ± 0.015**	**0.391 ± 0.062**	**0.0018 ± 0.0001**
Hybrid ensemble (+VQE)	0.822 ± 0.009	0.469 ± 0.037	0.0020 ± 0.0001

With only two held-out targets and ten seeds each, the five-qubit experiment is not powered to resolve the ZNE-versus-noisy comparison. It is adequate to show that the learned pipeline does not beat the noisy baseline at five qubits, which is the claim the paper makes. The five-qubit training set is proportionally smaller than the four-qubit one, so the five-qubit result also reflects data scarcity rather than a clean test of scaling.

6.5 Summary of findings

Taken together, the results support three claims. First, once the label-degeneracy and leakage defects are fixed, the learned pipeline does not outperform the unmitigated output on fidelity at four

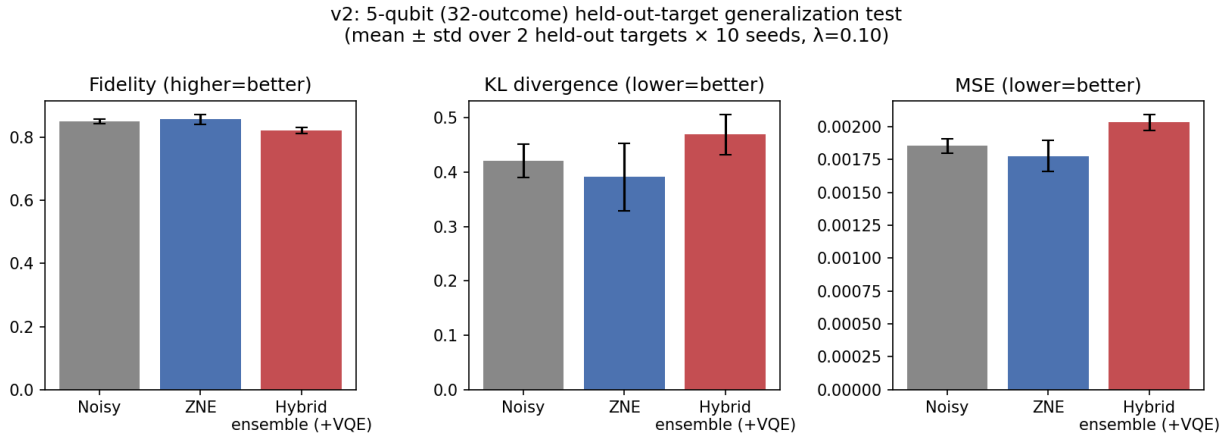


Figure 3: 5-qubit generalization check (mean ± standard deviation over 2 targets × 10 seeds).

qubits, and it trails ZNE on every metric. Second, the ordering holds for each held-out target and at five qubits, although the five-qubit margin is small. Third, the VQE feature does not improve the learned pipeline under this training budget.

7 Discussion

7.1 What the defects imply for learned mitigation

Label degeneracy and label leakage are simple to introduce and easy to miss. The degeneracy was visible only after counting distinct labels, and the leakage was visible only after reading the ensemble formula closely. Any learned mitigation method that trains on one circuit with a fixed seed inherits the same problem, and any method whose evaluation uses the ideal output at inference, even indirectly, inherits the leakage. We recommend that learned-mitigation studies report the number of distinct training labels, evaluate on targets absent from training, use a leak-free weighting or selection rule, and report seed-level variability. These checks are inexpensive relative to the cost of a misleading result.

7.2 What the negative result does and does not show

The corrected experiments do not prove that learned error mitigation cannot work. They show that this pipeline, with these features, these models, and this training corpus, does not beat ZNE. A larger corpus, richer features, or a model class that exploits the structure of the ideal output (for example, a permutation-aware model, since the ideal Grover output depends deterministically on the marked bitstring) might change the outcome. We did not test those alternatives, and we do not claim they would succeed.

7.3 Why the learned pipeline may underperform the noisy baseline

That the learned pipeline is worse than no mitigation on fidelity is the most striking result. One plausible explanation is that the models are trained to regress the ideal vector from a 16-dimensional histogram and a few scalars, and with about 900 samples they learn a smooth average that is not sharp enough to reproduce the marked-state peak. The regressors then produce outputs whose fidelity to the ideal is lower than the noisy input's. This explanation is consistent with the higher KL and MSE of the learned variants, and it would predict that more data or a sharper output model would close the gap. We did not test that prediction.

7.4 Why ZNE is strong here

ZNE is a simple, low-parameter method, and in this noise model its extrapolation is well behaved, because the noise is applied uniformly to every gate. It requires no training data and no assumption about the target. Its strength here should not be read as a general result, since real devices have structured, drifting noise. Its strength under the simulator model is consistent with the view that model-based mitigation is a strong baseline that learned methods must beat under a fair protocol.

7.5 Interpreting the five-qubit result

The five-qubit check shows the same direction as the four-qubit result, with a smaller margin and fewer targets. A scaling claim would require more targets, more seeds, and a training budget that scales with the output dimension. We therefore make no claim about how the gap evolves with qubit count beyond the observation that it does not favour the learned pipeline at either size.

8 Threats to Validity

Noise model. The noise is a simulator model of depolarizing and amplitude damping at a single level per circuit. It omits readout error, two-qubit gate error, crosstalk, and drift. Results may differ on hardware, and the ranking of ZNE and learned methods could change under structured noise.

ZNE implementation. The baseline uses a two-point linear extrapolation, and the third scale is unused. A more careful ZNE, with higher-order fits and all three scales, could perform better or worse. Because ZNE is the strongest method in our comparison, the choice of implementation matters for the conclusion, and we would welcome a comparison with stronger extrapolation methods.

Target coverage. The four-qubit test uses three targets and the five-qubit test uses two. Target-specific effects are visible (Table 2), so three targets do not characterize the space of marked states.

Training budget. The training set has 900 samples for a 16-dimensional label at four qubits and 240 samples for a 32-dimensional label at five. The learned pipeline may be underpowered at both sizes, and the negative result should be read as a result for this budget.

Model and feature choices. We tested one RF, one MLP, and one VQE probe configuration. Other regressors, other probes, and other feature representations could change the learned results, and we have not explored them systematically.

Statistical scope. The reported standard deviations are over shot-noise seeds for a fixed set of targets. They do not capture variability over training-set construction or model initialization beyond seed 42.

9 Limitations

- Only single-marked-target Grover circuits are studied. Multi-marked and structured circuits are not covered.
- The noise model is a simulator model, not hardware calibration.
- The held-out evaluation uses three targets (four qubits) and two (five qubits).
- The VQE optimizer (COBYLA, 50 iterations with random restarts) is simple and may limit the feature's quality.
- The five-qubit experiment has fewer training samples relative to output dimension.
- The ZNE baseline is a two-point linear extrapolation.

10 Reproducibility

The revised pipeline, trial-level records, aggregated tables, plots, and exact configurations are in the public repository [6]. The four-qubit experiment runs with `python3 main_v2.py` in about three minutes, and the five-qubit experiment runs with `python3 scaling_v2.py` in about one minute. Each run writes a timestamped folder under `results/` containing `raw_eval_records.csv`, `summary_per_target_mean_std.csv`, `summary_pooled_mean_std.csv`, a bar chart, and `config_used.json`. Targets are bitstrings with leading zeros, so `raw_eval_records.csv` must be loaded with `dtype={"target": str}`, because the default CSV parser otherwise drops the leading zero of `0100`. The original submission is preserved unmodified under `v1_original/` for side-by-side comparison. Dependencies are listed in `requirements.txt`; the VQE probe depends on the EstimatorV2 primitive in Qiskit Aer, so the exact Qiskit version should match the one recorded in the repository.

The figures in this paper are copies of the bar charts saved with the 4-qubit and 5-qubit runs. All tables are computed from the saved summary CSVs.

11 Conclusion

A hybrid ML+VQE pipeline that reported large gains in an earlier version does not hold up under a corrected evaluation. The defects we identify, constant training labels, a leaking ensemble weight, and a noise-independent probe feature, are general pitfalls for learned error mitigation. Under

the corrected held-out protocol, zero-noise extrapolation outperforms the learned pipeline at four and five qubits, and at four qubits the learned pipeline is worse than the unmitigated output on fidelity. We present the corrected evaluation as the contribution, and we recommend that future learned-mitigation work adopt a held-out multi-target protocol, a leak-free weighting rule, and seed-level reporting as standard practice.

A Configuration

Table 4: Four-qubit experiment configuration (`config.py` and `config_used.json`).

Parameter	Value
Qubits	4
Shots per execution	2048
Training targets	10 (see Section 5)
Held-out targets	`1011`, `0100`, `1101`
Training noise levels	six values in $[0.02, 0.18]$
Repetitions per cell	15
Training matrix shape	900×18
Test noise level	$\lambda = 0.10$
Evaluation seeds per target	20
Global seed	42
RF	200 trees
MLP	hidden $(128, 64, 32)$, early stopping, 2,000 max iterations
ZNE scales	$0.05, 0.10, 0.15$ (two-point linear extrapolation)
VQE probe	`TwoLocal`(ry, rz; cz; 2 reps), COBYLA, 50 iterations, 1,024 shots

Table 5: Five-qubit experiment configuration (`config.py`).

Parameter	Value
Qubits	5 (32 outcomes)
Training targets	`00000`, `01010`, `10101`, `11111`, `00110`, `11001`
Held-out targets	`10011`, `01101`
Training noise levels	four values in $[0.02, 0.18]$
Repetitions per cell	10
Test noise level	$\lambda = 0.10$
Evaluation seeds per target	10

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